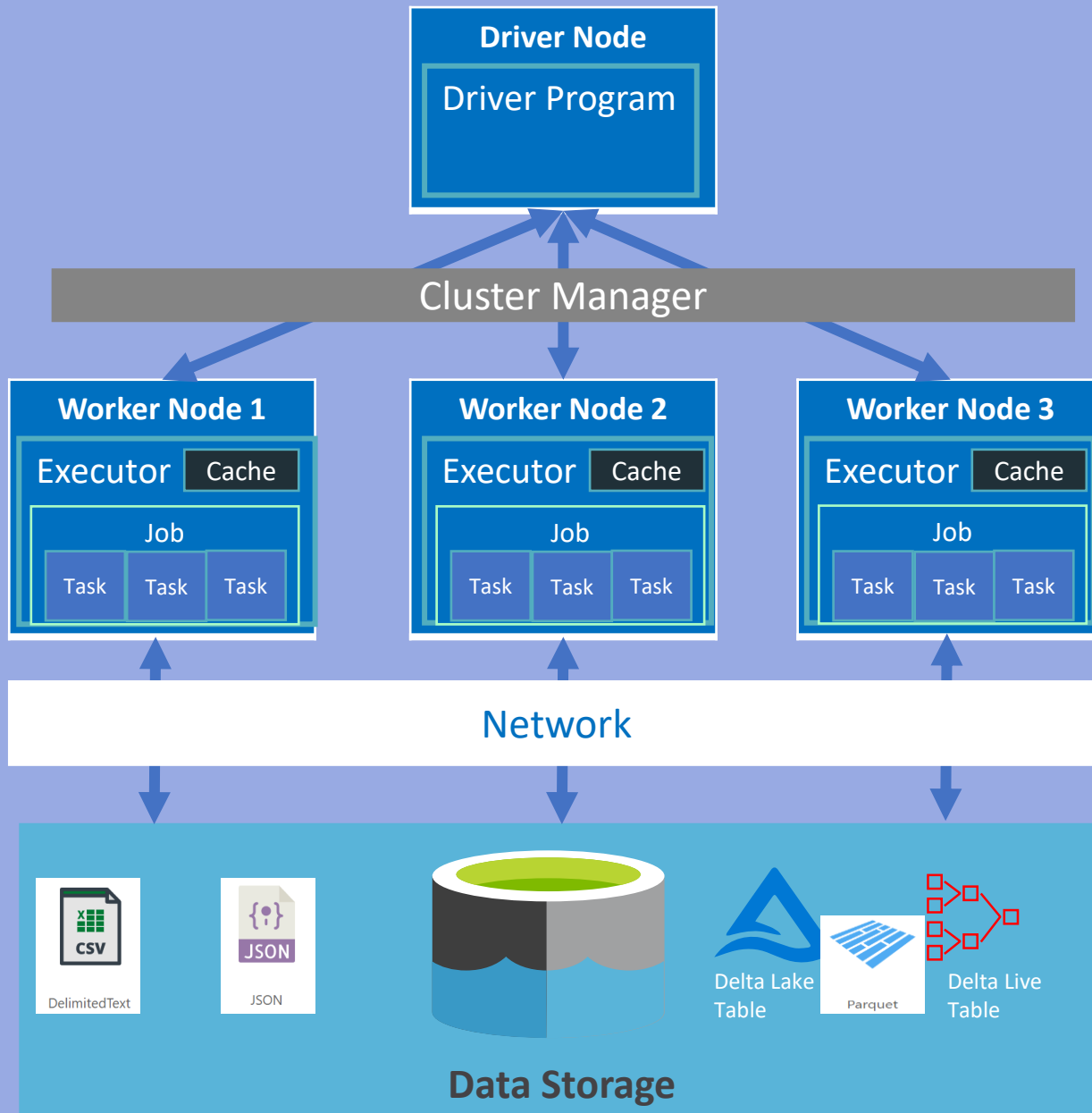


Performance Tuning – Best Practices



SELECT only required columns from Table (**Column Pruning**)

FILTER SOURCE DATA at the source tables(**PREDICATE Pushdown**)

SELECT specific partition of the data (**PARTITION Pruning**)

DELTA Cache – Creating Copies of remote files data in nodes (Using SSD)
`set spark.databricks.io.cache.enabled = true`

Spark Cache – use `cache()` or `persist()` keywords to keep the data in the cache until Spark Session ends , Useful when multiple ACTIONS on same Data frame

CREATE TEMP VIEWS for intermediate processing of data

Delta Merge – Use **MERGE** SQL Statements to combine insert/update/delete operations on same tables

Frequently run **VACUUM** statements to remove unused historical data files
Change default retention period threshold values to custom values
`deltaTable.deletedFileRetentionDuration = "interval 15 days"`

Always use **Job Cluster** for automated jobs , Define Job Cluster Config at Task Level

Choose **2-4 worker** nodes for small workloads and **8-10** for larger workloads

Use **Photon** enabled clusters and **Liquid Clustering** on Delta Lake Tables
Liquid clustering in Databricks dynamically organizes data for faster queries without fixed partitions